▷ **OLD IRON**
The banded iron formations (alternate layers of iron-rich deposits), exposed here in the Hamersley Range on one of Australia's cratons, are 2.5 billion years old.

▷ **ANCIENT RIVER**
The Finke River first started to flow over 350 million years ago. At that time, it flowed freely over the plain above its present-day gorge.

◁ **SMOKING ISLAND**
Mount Tavurvur is one of the most active vents on the highly volatile Rabaul caldera on New Britain, Papua New Guinea.

THE SHAPING OF AUSTRALIA AND NEW ZEALAND

Australia is a flat, time-worn continent. It holds some of Earth's oldest rocks, and its remarkable landscapes have been carved out over hundreds of millions of years.

Ancient crust

The western half of the continent is the Australian Shield of old crustal rock, built around three ancient sections of stable crust called cratons. Zircon crystals in the Jack Hills, discovered in rocks on one of the cratons, are 4.4 billion years old—the oldest datable material on Earth. Another craton is, along with South Africa's Kaapvaal Craton, one of only two unaltered chunks of Archean crust on Earth. The Australian Alps formed over 360 million years ago—as a result, they have been worn lower than other ranges and have no jagged peaks. Instead, they are made of plateaus and gorges.

Australia has the **most ancient landscape** and holds **Earth's oldest rocks**

Volcanic islands

Along its northern edge, Australia has crunched into a complex melee of small tectonic plates to the north of Papua New Guinea. There are more than five small plates here, including the South and North Bismarck Plates under New Britain. These small plates jostle to and fro as they are squeezed between the giant Australian and Pacific Plates converging from either side. Some of the small plates will not last long because they are subducting beneath other plates and melting into Earth's mantle. Molten magma from the mantle bursts through the overriding plates in a series of volcanoes to make the islands of Melanesia—the most volcanically active region in the world. There are 22 volcanoes here that have erupted to form islands, while others erupt under the sea. The Rabaul caldera on the northern tip of New Britain contains several violently explosive vents.

Solo journey

Australia was attached to India and Antarctica as part of the great southern continent of Gondwana 150 million years ago, but about 130 million years ago a rift opened inside Gondwana and the three landmasses split away. About 100 million years ago, India broke

KEY EVENTS

450 million years ago An ancient ocean forms where Australia will eventually be. The sandstone seafloor will later be pushed upward by tectonic movement—both Uluru and Mount Olga are remnants of this seafloor.

300–280 million years ago The Great Dividing Range forms when Australia collides with landmasses that will become South America and Zealandia.

180 million years ago Rain forests arise that will eventually survive to become the Gondwanan rain forests in southeastern Australia.

85 million years ago A landmass that will eventually fall below sea level and become Zealandia splits away from Australia while it is still attached to Gondwana.

80 million years ago A landmass that will become Australia and Papua New Guinea breaks away from the rest of Gondwana.

50 million years ago Australia and Papua New Guinea break away from Antarctica and drift north, while Antarctica drifts southward to the South Pole.

25 million years ago What was once an ancient seafloor thrusts upward due to tectonic activity and becomes the Nullarbor Plain.

500,000 years ago The Great Barrier Reef first forms, but it has receded and expanded since then.

6,000 years ago New Guinea becomes an island as sea levels rise after the ice age.